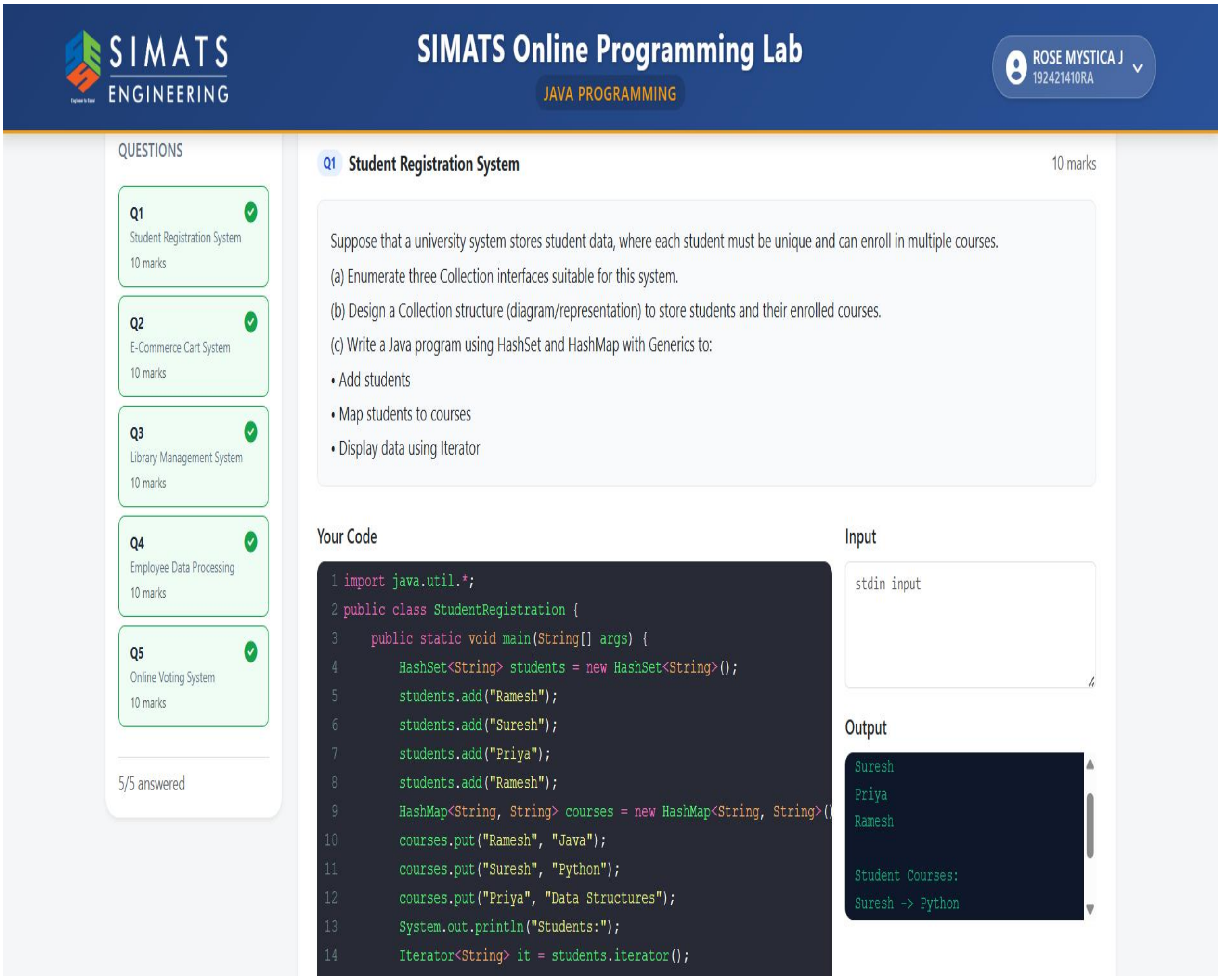
SIMATS ENGINEERING

CO2-AT2-SCENARIO BASED ASSESSMENT

COURSE CODE: CSA0904 **NAME:** Rose Mystica. J

COURSE NAME: Programming in JAVA **REG NO:** 192421410



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JAVA PROGRAMMING

ROSE MYSTICA J
192421410RA

QUESTIONS

Q1 Student Registration System 10 marks

Q2 E-Commerce Cart System 10 marks

Q3 Library Management System 10 marks

Q4 Employee Data Processing 10 marks

Q5 Online Voting System 10 marks

5/5 answered

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
import java.util.*;
public class ShoppingCart {
    public static void main(String[] args) {
        ArrayList<String> cart = new ArrayList<String>();
        cart.add("Laptop");
        cart.add("Mobile");
        cart.add("Headphones");
        cart.add("Mouse");
        System.out.println("Cart Items:");
        Iterator<String> it = cart.iterator();
        while (it.hasNext()) {
            System.out.println(it.next());
        }
        cart.remove("Headphones");
```

Input

stdin input

Output

```
Cart Items:
Laptop
Mobile
Headphones
Mouse
```

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5/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```
import java.util.*;
public class LibraryManagement {
    public static void main(String[] args) {
        HashMap<String, String> books = new HashMap<String, String>();
        books.put("ISBN101", "Java Programming");
        books.put("ISBN102", "Data Structures");
        books.put("ISBN103", "Computer Networks");
        System.out.println("Book Retrieved:");
        System.out.println("ISBN102 -> " + books.get("ISBN102"));
        System.out.println("\nAll Books:");
        for (Map.Entry<String, String> entry : books.entrySet()) {
            System.out.println(entry.getKey() + " -> " + entry.getValue
        }
    }
```

Input

stdin input

Output

```
Book Retrieved:
ISBN102 -> Data Structures

All Books:
ISBN103 -> Computer Networks
ISBN102 -> Data Structures
```

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5/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

Your Code

```java
import java.util.*;
public class Employee {
    public static void main(String[] args) {
        List<String> names = new ArrayList<String>();
        List<Double> salaries = new ArrayList<Double>();
        names.add("Ramesh");
        salaries.add(45000.0);
        names.add("Suresh");
        salaries.add(60000.0);
        names.add("Priya");
        salaries.add(55000.0);
        names.add("Anu");
        salaries.add(40000.0);
        System.out.println("All Employees:");
        Iterator<String> it = names.iterator();
```

Input

stdin input

Output

```
All Employees:
Ramesh - 45000.0
Suresh - 60000.0
Priya - 55000.0
Anu - 40000.0
```

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- Q5 Online Voting System — 10 marks

5/5 answered

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting • Display results

Your Code

```java
import java.util.*;
public class OnlineVoting {
    public static void main(String[] args) {
        HashSet<String> voters = new HashSet<String>();
        HashMap<String, Integer> votes = new HashMap<String, Integer>(
        votes.put("Candidate A", 0);
        votes.put("Candidate B", 0);
        votes.put("Candidate C", 0);
        String[] voterNames = {
            "Ramesh", "Suresh", "Priya", "Anu", "Ramesh"
        };
        String[] choices = {
            "Candidate A", "Candidate B", "Candidate A",
            "Candidate C", "Candidate B"
        };
```

Input

stdin input

Output

```
Ramesh voted for Candidate A
Suresh voted for Candidate B
Priya voted for Candidate A
Anu voted for Candidate C
Ramesh - Duplicate vote rejected
```